



LITERATURE REVIEW

Evaluating Recommender Systems under Sparse User Feedback

Kamal Lalloo
2652159

School of Computer Science and Applied Mathematics
Faculty of Science
University of the Witwatersrand

Supervisor: Hairong Bau

April 19, 2026

Chapter 1

Introduction

Recommender systems are widely used in modern online platforms to provide users with personalized suggestions, such as products, movies, and news content. As a key area within machine learning and information retrieval, their primary goal is to predict user preferences from historical interaction data. A widely adopted approach is collaborative filtering, which identifies patterns in user–item interactions. Early methods, including item based collaborative filtering and matrix factorization, have been effective by modeling similarities between items or learning latent representations of users and items (Sarwar et al., 2001; Koren et al., 2009).

Despite their success, these methods depend strongly on the availability and quality of user feedback. In many real-world settings, explicit feedback such as ratings is limited, and systems instead depend on implicit signals such as clicks or purchases. This data is often noisy, incomplete, and imbalanced (Hu et al., 2008). To address this, ranking-based methods such as Bayesian Personalized Ranking (BPR) were introduced, focusing on ordering relevant items rather than predicting exact preference scores (Rendle et al., 2009). More recent approaches, including deep learning models, aim to capture complex, non-linear user–item relationships (He et al., 2017; Liang et al., 2018), while sequential recommendation methods model the temporal dynamics of user behavior using architectures such as recurrent neural networks and transformers (Hidasi et al., 2016; Kang and McAuley, 2018; Sun et al., 2019).

Despite these advances, a fundamental challenge remains: data sparsity. In practice, users engage with only a tiny portion of the items, which limits the information available for learning reliable patterns. This challenge is further intensified by the cold-start problem, where newly introduced users or items lack sufficient interaction history, and by biases such as the over-representation of popular items (Abdollahpouri et al., 2019). Although approaches such as graph-based models and self-supervised learning attempt to mitigate these limitations (Wu et al., 2021), their effectiveness under varying levels of sparsity is not fully understood.

This leads to the central research problem: how robust are different recommender system approaches when interaction data is sparse? In particular, it remains unclear how classical methods compare to more complex modern models under constrained data conditions.

This literature review addresses this problem by examining existing recommender system approaches in the context of data sparsity. It first provides background on foundational methods, followed by a review of modern approaches including deep learning, sequential, and graph-based models. It then discusses techniques for handling sparse data and evaluates the limitations of current methods. Through this analysis, the review aims to identify key gaps in the literature and motivate the need for a systematic evaluation of recommender systems under controlled sparsity conditions.

1.1 Problem Statement

A key objective of recommender systems is to learn accurate and reliable user preferences from interaction data. In practice, however, these systems operate under highly sparse conditions, where users interact with only a small fraction of available items. This limits the information available for learning and often leads to reduced recommendation quality, particularly for new or infrequent users. The cold-start problem further intensifies this issue, as new users and items lack sufficient interaction data for meaningful predictions (Glisovic et al., 2025). In addition, reliance on implicit feedback introduces noise and uncertainty, making it more difficult to distinguish between relevant and irrelevant items.

Although many models have been proposed to address these challenges, there remains limited systematic understanding of how different approaches perform under varying levels of data sparsity. Prior work shows that more complex models do not consistently outperform simpler methods, with well-tuned classical approaches sometimes achieving comparable results (Ferrari Dacrema et al., 2019). Furthermore, recent studies highlight that models can be sensitive to missing or incomplete interactions, raising concerns about their robustness in realistic settings (Siciliano et al., 2024). These limitations motivate the need for a systematic evaluation of recommender systems under controlled sparsity conditions to better understand their performance and robustness.

Chapter 2

Background and Related Work

2.1 Background

Recommender systems aim to infer user preferences by identifying patterns in past interaction data. This data is commonly represented as a user–item interaction matrix, where each entry indicates whether a user has engaged with a particular item, such as through clicks, purchases, or ratings. In practice, this matrix is usually sparse, since users engage with only a limited subset of available items, making it harder to learn reliable patterns from the data (Koren et al., 2009).

A foundational approach to recommendation is collaborative filtering, which is based on the idea that users with similar past behavior are likely to share similar preferences in the future. Early methods, such as item-based collaborative filtering, compute similarities between items based on co-occurrence patterns and recommend items that resemble those a user has previously interacted with (Sarwar et al., 2001). These methods depend on observed interaction data to estimate similarities between users or items.

To model user preferences more effectively, matrix factorization techniques were introduced, which learn low-dimensional latent representations of users and items. These models project users and items into a shared latent space, where interactions are typically modeled using operations such as inner products (Koren et al., 2009). Matrix factorization is widely used in recommender systems and is often treated as a baseline in the literature.

In many real-world applications, recommender systems depend on implicit feedback, such as clicks or purchases, instead of explicit ratings. Unlike explicit feedback, implicit data does not directly indicate preference strength and typically lacks clearly defined negative examples, which makes it more challenging to interpret (Hu et al., 2008). To address this, ranking-based methods such as Bayesian Personalized Ranking (BPR) were introduced, which learn by encouraging observed interactions to be ranked above unobserved ones rather than predicting exact preference scores. (Rendle et al., 2009).

These foundational methods form the basis of many recommender systems and have been widely studied in the literature. However, their performance is influenced by the characteristics of the available interaction data, which has motivated the development of more advanced models that aim to capture more complex relationships and improve recommendation quality.

2.2 Related Work

Building on the foundational methods discussed in the previous section, a wide range of recommender system models have been proposed to address the limitations of traditional approaches. These developments include deep learning models that aim to capture more complex user–item relationships, sequential methods that incorporate temporal dynamics, and graph-based approaches that model higher-order interactions. In parallel, other lines of work have focused on improving learning under limited data conditions through techniques such as self-supervised learning and data augmentation. Despite these advances, challenges related to data sparsity and robustness remain, motivating continued research into the behavior of these models under realistic conditions.

2.2.1 Deep Learning for Recommendation

To address the limitations of linear models such as matrix factorization, recent work has explored deep learning approaches that aim to model complex, non-linear relationships between users and items. One of the early examples is Neural Collaborative Filtering, which replaces the standard dot product interaction with a neural network, enabling the model to learn more flexible interaction patterns (He et al., 2017).

Another important direction is the use of probabilistic deep models, such as variational autoencoders (VAEs), which represent user preferences as latent variables and learn distributions over user behavior (Liang et al., 2018). These models extend traditional latent factor approaches by introducing non-linear generative processes and sharing statistical strength across users, which has been explored as a way to improve robustness in sparse settings. More broadly, deep learning techniques, including multilayer perceptrons, recurrent networks, and autoencoders, have been applied in recommender systems to learn richer representations and capture complex data patterns (Zhang et al., 2019).

While deep learning models provide more flexible function approximations than classical approaches, they do not fully resolve the challenges associated with sparse data. In many cases, their performance remains sensitive to the availability of interaction data, and they can be prone to overfitting due to their large number of parameters. As a result, although deep learning has advanced recommender system modeling, it does not fully solve the challenges associated with data sparsity.

2.2.2 Sequential Recommendation Models

Traditional recommender systems often model user preferences without explicitly considering the order of interactions. To address this, sequential recommendation models were introduced, which aim to predict future user behavior based on interaction sequences. These models represent user activity as ordered sequences and capture dependencies between successive interactions, making them suitable for applications where preferences evolve over time.

One of the earlier approaches in this area uses recurrent neural networks (RNNs) to model user sessions. Session-based RNN models process sequences of interactions within a session and use

this information to generate recommendations, particularly in cases where long-term user profiles are unavailable (Hidasi et al., 2016). These models maintain a hidden state that summarizes past interactions and use it to predict future interactions based on the session history. Building on this idea, more recent methods adopt self-attention mechanisms. For example, SASRec uses attention to identify and weigh relevant past interactions when predicting the next item (Kang and McAuley, 2018). Similarly, BERT4Rec extends this approach by using bidirectional self-attention, enabling the model to learn representations from both past and future context within a sequence (Sun et al., 2019).

While these models provide a framework for capturing user behavior over time, they do not eliminate the challenges associated with limited interaction data. Their performance can be affected by sparse or incomplete interaction histories, as shorter or less informative sequences may provide limited context for learning dependencies. As a result, handling sparsity remains an important consideration for sequential recommendation models in practical settings.

2.2.3 Graph-Based Recommendation

Graph-based recommender systems extend traditional approaches by representing user–item interactions as a graph. In this formulation, users and items are treated as nodes, while their interactions form edges in a bipartite graph. This representation allows models to capture relationships beyond direct interactions by considering connections through shared neighbors, incorporating higher-order interaction patterns (Wang et al., 2019).

One of the key models in this area is Neural Graph Collaborative Filtering (NGCF), which applies graph neural networks to propagate embeddings across the user–item interaction graph. Through iterative message passing, NGCF aggregates information from multi-hop neighbors, allowing embeddings to encode higher-order relationships between items and users (Wang et al., 2019). Building on this approach, LightGCN simplifies the graph convolution framework by focusing on neighborhood aggregation and removing feature transformations and nonlinear activations. This design reduces model complexity and has been shown to achieve competitive performance in recommendation tasks (He et al., 2020).

Further developments focus on improving representation learning through additional training objectives. For example, Neural Contrastive Learning (NCL) introduces self-supervised learning techniques that leverage structural and semantic relationships within the interaction graph, with the aim of improving learned representations under limited data conditions (Lin et al., 2022).

While graph-based approaches provide a framework for modeling complex relationships in user–item interactions, they remain dependent on the structure and quality of the underlying interaction data. Since the graph is constructed from observed interactions, sparse or incomplete data can limit the available connectivity and influence the information that can be propagated through the network. As a result, graph-based models do not fully eliminate the challenges associated with sparse and noisy interaction data.

2.2.4 Handling Sparse Data and Cold Start

Data sparsity and cold-start scenarios remain key challenges in recommender systems, particularly when users or items have limited interaction history. To address these issues, a range of methods has been proposed to support learning under limited data conditions. Broadly, these approaches focus on incorporating additional information or generating additional training signals from existing data.

One common strategy is to combine collaborative filtering with auxiliary information. For example, DropoutNet simulates missing interactions during training using dropout, with the aim of improving performance in cold-start scenarios by incorporating both interaction data and side information (Volkovs et al., 2017). Other approaches focus on data augmentation. Generative feedback models, for instance, simulate user interactions by creating synthetic feedback, providing additional training signals for learning from limited data (Wang et al., 2021).

In addition to model-specific approaches, survey work has examined broader categories of cold-start solutions, including content-based, collaborative, and hybrid methods (Glisovic et al., 2025).

Despite the range of proposed approaches, no single method has been shown to fully address the challenges associated with sparse data and cold-start scenarios. The effectiveness of different methods depends on factors such as the availability of additional information, the structure of the data, and the application context. As a result, handling sparsity remains an open area of research in recommender systems.

2.2.5 Self-Supervised and Contrastive Learning

Recent work has explored self-supervised and contrastive learning as an approach to learning from limited interaction data. Rather than relying solely on observed user feedback, these methods generate additional training signals directly from the data, allowing models to learn from augmented or transformed views of the original interactions.

In sequential recommendation, S^3 -Rec introduces self-supervised pretraining tasks based on mutual information maximization to capture relationships between items, attributes, and sequences before optimizing the main recommendation objective (Zhou et al., 2020). Similarly, CL4SRec applies contrastive learning by constructing augmented versions of user interaction sequences, such as through cropping, masking, and reordering, and training the model to distinguish between similar and dissimilar sequence views (Xie et al., 2021).

This idea has also been applied to graph-based recommendation models. For example, Self-supervised Graph Learning (SGL) creates multiple variations of the interaction graph such as by randomly removing nodes or edges, or using random walks and then learns from these variations using contrastive learning (Wu et al., 2021). Building on this, Neighborhood-enriched Contrastive Learning (NCL) incorporates both structural relationships from the graph and similarity between learned representations into the training objective (Lin et al., 2022).

These methods introduce additional training signals that can be derived from the structure and patterns within the data. However, they do not eliminate the challenges associated with sparse and incomplete interaction data, and their effectiveness depends on factors such as the data characteristics and the choice of augmentation or training strategy.

2.2.6 Robustness and Real-World Challenges

While many recommender models perform well under controlled conditions, their effectiveness can be affected in real-world settings due to missing data, noise, and bias. In practice, user interactions are often incomplete, inconsistent, and sometimes unreliable, which can impact model behavior. This highlights the importance of designing systems that are robust to imperfect data.

One key issue is the sensitivity of models to missing interactions, particularly in sequential recommendation. These models rely on patterns in user interaction sequences, and missing elements in a sequence can affect model predictions. Recent work shows that even small perturbations, such as removing individual interactions, can influence model outputs, as sequential recommenders can depend on recent items in a sequence (Siciliano et al., 2024). Robust training approaches attempt to address this by modifying the learning process to account for missing inputs, with the aim of improving prediction stability. Similarly, models such as RETR focus on identifying more informative user behaviors through mechanisms such as pathway attention, which emphasizes relevant interactions while reducing the influence of less informative ones (Yao et al., 2022).

Noise in interaction data is another challenge. User actions such as accidental clicks or inconsistent behavior can introduce misleading signals into the learning process. Approaches such as graph-based filtering or pruning techniques attempt to reduce the influence of less informative interactions, with the aim of improving model stability.

In addition to missing data and noise, recommender systems are also affected by popularity bias, where frequently interacted items are recommended more often than less popular ones. This can limit diversity and reinforce imbalances in the data, particularly in sparse settings where popular items dominate the interaction space. Prior work shows that recommender systems tend to over-represent popular items while under-representing long-tail items, which can reduce item discovery (Abdollahpouri et al., 2019). Methods addressing this issue often apply re-ranking strategies to balance accuracy with diversity and improve coverage of less frequent items.

Overall, these challenges indicate that real-world recommendation is influenced not only by model design but also by the quality of the underlying data. Handling missing data, reducing noise, and mitigating bias are important considerations in developing reliable recommender systems.

2.2.7 Critical Perspectives and Model Comparison

Despite the rapid development of increasingly complex recommender system models, recent work has questioned whether these advances consistently lead to improved performance. In particular,

concerns have been raised regarding evaluation practices in the literature and the extent to which deep learning approaches provide benefits over simpler methods.

A notable study by Dacrema et al. examines the performance of deep learning-based recommender models under consistent evaluation settings and reports that many do not consistently outperform well-tuned classical baselines (Ferrari Dacrema et al., 2019). Their analysis shows that, among reproducible models, several deep learning approaches perform comparably to simpler methods such as nearest-neighbor and graph-based models, and in some cases even non-personalized methods are competitive. The authors attribute these findings to factors such as weak baselines, inconsistent experimental setups, and limited reproducibility, suggesting that reported improvements may not always reflect meaningful advances.

Similarly, the EASE-R model provides an example of a simple linear approach that achieves competitive performance relative to more complex methods (Steck, 2019). Despite its simplicity, EASE-R has been reported to perform comparably to, and in some cases outperform, more complex collaborative filtering models on standard benchmarks. These findings indicate that increased model complexity does not necessarily lead to better results.

Overall, this line of work highlights the importance of careful evaluation, strong baselines, and consideration of data characteristics when comparing recommender system models. In particular, these considerations are relevant in sparse data settings, where the relative performance of different approaches may vary depending on the availability and structure of the data.

2.3 Summary

This chapter reviewed the development of recommender systems, from classical collaborative filtering methods to modern deep learning, sequential, and graph-based approaches. While classical approaches such as item-based filtering and matrix factorization remain strong baselines, their performance is limited in sparse data settings. More advanced models improve expressiveness by capturing non-linear relationships and temporal dynamics, but they remain dependent on sufficient interaction data.

Recent work has focused on mitigating sparsity through graph-based learning, self-supervised techniques, and cold-start strategies, aiming to extract more meaningful representations from limited data. However, challenges such as missing interactions, noise, and popularity bias continue to impact performance and generalization in real-world systems.

Importantly, prior studies show that increased model complexity does not necessarily lead to better performance, with simpler models often remaining competitive when properly evaluated (Ferrari Dacrema et al., 2019; Steck, 2019).

Overall, despite significant progress, the behavior of recommender systems under varying levels of data sparsity is not fully understood. This highlights the need for systematic evaluation to better assess model robustness and guide practical system design.

Bibliography

- [Abdollahpouri et al., 2019] Abdollahpouri, H., Burke, R., and Mobasher, B. (2019). Managing popularity bias in recommender systems with personalized re-ranking. In *FLAIRS Conference*.
- [Ferrari Dacrema et al., 2019] Ferrari Dacrema, M., Cremonesi, P., and Jannach, D. (2019). Are we really making much progress? a worrying analysis of recent neural recommendation approaches. In *ACM Conference on Recommender Systems (RecSys)*, pages 101–109.
- [Glisovic et al., 2025] Glisovic, N., Kragic, D., and Tegner, M. (2025). Item cold start in e-commerce recommender systems: A survey. *IEEE Access*, 13.
- [He et al., 2020] He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., and Wang, M. (2020). Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference*, pages 639–648.
- [He et al., 2017] He, X., Liao, L., Zhang, H., Nie, L., Hu, X., and Chua, T.-S. (2017). Neural collaborative filtering. In *Proceedings of the 26th International World Wide Web Conference (WWW)*, pages 173–182.
- [Hidasi et al., 2016] Hidasi, B., Karatzoglou, A., Baltrunas, L., and Tikk, D. (2016). Session-based recommendations with recurrent neural networks. In *International Conference on Learning Representations (ICLR)*.
- [Hu et al., 2008] Hu, Y., Koren, Y., and Volinsky, C. (2008). Collaborative filtering for implicit feedback datasets. In *IEEE International Conference on Data Mining (ICDM)*, pages 263–272.
- [Kang and McAuley, 2018] Kang, W.-C. and McAuley, J. (2018). Self-attentive sequential recommendation. In *IEEE International Conference on Data Mining (ICDM)*, pages 197–206.
- [Koren et al., 2009] Koren, Y., Bell, R., and Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *IEEE Computer*, 42(8):30–37.

- [Liang et al., 2018] Liang, D., Krishnan, R. G., Hoffman, M. D., and Jebara, T. (2018). Variational autoencoders for collaborative filtering. In *World Wide Web Conference (WWW)*, pages 689–698.
- [Lin et al., 2022] Lin, Z., Tian, C., Hou, Y., and Zhao, W. X. (2022). Improving graph collaborative filtering with neighborhood-enriched contrastive learning. In *ACM Web Conference (WWW)*.
- [Rendle et al., 2009] Rendle, S., Freudenthaler, C., Gantner, Z., and Schmidt-Thieme, L. (2009). Bpr: Bayesian personalized ranking from implicit feedback. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 452–461.
- [Sarwar et al., 2001] Sarwar, B., Karypis, G., Konstan, J., and Riedl, J. (2001). Item-based collaborative filtering recommendation algorithms. In *World Wide Web Conference (WWW)*, pages 285–295.
- [Siciliano et al., 2024] Siciliano, F., Lagziel, S., Gamzu, I., and Tolomei, G. (2024). Robust training of sequential recommender systems with missing input data. In *RobustRecSys Workshop at RecSys*.
- [Steck, 2019] Steck, H. (2019). Embarrassingly shallow autoencoders for sparse data. In *World Wide Web Conference (WWW)*.
- [Sun et al., 2019] Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., and Jiang, P. (2019). Bert4rec: Sequential recommendation with bidirectional encoder representations from transformer. In *ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1441–1450.
- [Volkovs et al., 2017] Volkovs, M., Yu, G., and Poutanen, T. (2017). Dropoutnet: Addressing cold start in recommender systems. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [Wang et al., 2021] Wang, W., Xu, H., Zhang, R., Wang, W., Rai, P., and Carin, L. (2021). Learning to recommend from sparse data via generative user feedback. In *AAAI Conference on Artificial Intelligence*.
- [Wang et al., 2019] Wang, X., He, X., Wang, M., Feng, F., and Chua, T.-S. (2019). Neural graph collaborative filtering. In *ACM SIGIR Conference*, pages 165–174.
- [Wu et al., 2021] Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., and Xie, X. (2021). Self-supervised graph learning for recommendation. In *ACM SIGIR Conference*, pages 726–735.

- [Xie et al., 2021] Xie, X., Sun, F., Liu, Z., Wu, S., Gao, J., Ding, B., and Cui, B. (2021). Contrastive learning for sequential recommendation. In *ACM SIGIR Conference*.
- [Yao et al., 2022] Yao, Z., Chen, X., Wang, S., Dai, Q., Li, Y., Zhu, T., and Long, M. (2022). Recommender transformers with behavior pathways. *arXiv preprint arXiv:2206.06804*.
- [Zhang et al., 2019] Zhang, S., Yao, L., Sun, A., and Tay, Y. (2019). Deep learning based recommender system: A survey and new perspectives. *ACM Computing Surveys*, 52(1).
- [Zhou et al., 2020] Zhou, K., Wang, H., Zhao, W. X., Zhu, Y., Wang, S., Zhang, F., Wang, Z., and Wen, J.-R. (2020). S³-rec: Self-supervised learning for sequential recommendation. In *ACM International Conference on Information and Knowledge Management (CIKM)*, pages 1893–1902.

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

- | | |
|---|-------------------------------------|
| 1. Idea Generation (research problem/design, hypothesis) | ☐ |
| 2. Sourcing Related Work (summarising, identifying sources) | ☒ |
| 3. Methods and Experiment Design (experiment setup, model tuning) | ☐ |
| 4. Data Analysis (presentation, coding, interpretation) | ☐ |
| 5. Theoretical Development (theorem proving, conceptual analysis) | ☐ |
| 6. Code Development (generating algorithms, writing scripts) | ☐ |
| 7. Presentation (rendering graphics, formatting) | ☐ |
| 8. Editing (grammar, readability) | ☒ |
| 9. Writing (text generation, document structuring) | ☒ |
| 10. Citation Formatting (structuring, organising) | ☒ |

If other uses were involved, please specify below:

Generative AI tool used (list all)	Used for?

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2652159

Candidate signature: 

Date: 19/04/2026